The effect of the relative permeability of steel armour on the inner screening factor

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***Abstract*—In today's world, as the power and sensitivity of electrical systems increase, addressing Electromagnetic Compatibility (EMC) becomes crucial. One such area of focus is electric traction and the signaling cables used for its safety devices. Although the proper shielding of these cables is essential in order to provide appropriate protection against induced voltage, a lot of cables with inappropriate shielding were found during the construction of railway lines. The aim of this paper is to reveal the causes behind the appropriate and inappropriate shielding. The methods also included measurements of relative permeability and material technology, like optical and electron microscopy. The results showed that the relative permeability of steel tape armour, which can be influenced by manufacturing techniques, is essential in order to provide a great screening factor.**

Keywords - signaling cable, screening factor, relative permeability, steel tape armour

I. Introduction

With the increasing traction currents in railway overhead lines today, it is essential to examine electromagnetic compatibility (EMC) in this environment. High alternating currents can induce voltages in signaling cables located near the railway infrastructure. [1] [2]

This problem raises the need for adequate cable protection against induced voltage. In this paper, we examine two coaxial cables, which consist of copper layer shielding and steel tape armoring. According to Lenz's law, current will flow in the steel tape armour and copper layer. This will flow in the opposite direction to the inducing current, thereby opposing the magnetic field and reducing its strength within the layer compared to the outside. This reduction is measured and expressed by the screening factor. [3]

In practice, significantly different screening factors have been observed in various railway signaling cables. However, the reasons for these discrepancies were not immediately apparent, as there were no visible signs of damage or irregularity. This prompted further investigation to identify the underlying causes.

This study compares two multicore railway signaling cables, both shielded with a copper layer and steel tape armour. Although the cables share identical structural designs and primary parameters (length, diameter of the cable and the cores, number of cores), a significant discrepancy was observed in their inner screening factors. The objective of this paper is to investigate and identify the underlying causes of these differences in screening performance.

II. The effect of relative permeability

A. Comparing the screening factors

The inner screening factors were measured according to [4], the results are shown below.

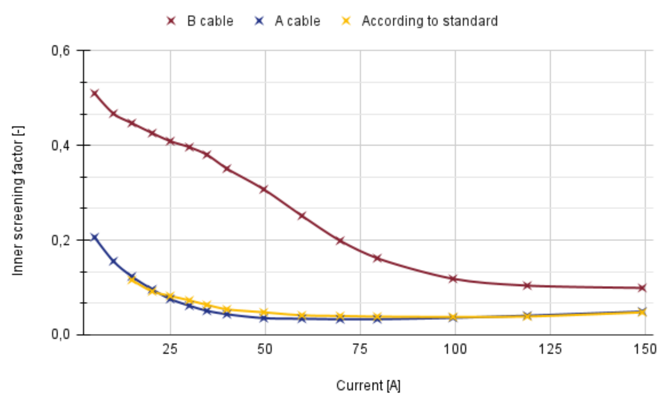

Fig. 1 — Screening factors

A significant difference can be observed between Cable A and Cable B. To understand this difference, let us consider the definition of the inner screening factor (1). This factor is calculated based on the inner and outer surface impedances [3].

$$|k_i| = \frac{|U_i|}{|U_o|} = \frac{|Z_i|}{|Z_o|} \; [-] \tag{1}$$

Where the inner surface impedance can be approximated by the value of the resistance component of the shielding (2) [3].

$$Z_i \cong R_i \cong R_{DC} \tag{2}$$

The value of the outer surface impedance can be approximated at low frequency by the DC resistance of the copper screening and the impedance from the steel armour, where the real part R_{armour}, while the imaginary part X_{armour} represents the significant inductive reactance within the armour (3) [3].

$$Z_o \cong R_{DC} + R_{armour} + X_{armour} \tag{3}$$

From (3), the R_{DC} and the resistance of the armour R_{armour} are constant, while the reactance of the armour is directly proportional to its inductance (4).

$$X_{armour} = j\omega L = \frac{U}{I} \tag{4}$$

The inductance of a coil depends only on the geometric factors and the value of the permeability [5]. In this case, the number of coil threads is $N = 1$.

$$L = \frac{\mu_0 \mu_r A N^2}{l} \, [H] \tag{5}$$

From geometrical parameters and from (4) and (5), the relative permeablity is definable as shown in (6) [3].

$$\mu_r = \frac{U}{j\omega I \mu_0 \frac{Q_{FE}}{\pi D_m}} [-] \tag{6}$$

From (6), it is evident that the relative permeability is a complex quantity; therefore, both voltage and current should also be measured in complex form. Q_{FE} can be calculated as $a \cdot b \cdot n$, where n is the total number of steel tapes per unit length, $a \cdot b$ is the cross-section of the steel tape. D_m is the diameter of the steel tape armour [3].

In summary, the inner screening factor is strongly influenced by the relative permeability, exhibiting an inverse proportional relationship $k_i \sim \frac{1}{\mu_r}$. Consequently, we began investigating the behavior of relative permeability, examining its values to determine whether a correlation exists with the differences in inner screening factors discussed at the beginning of this paper.

B. Relative permeability

The relative permeability is obtained from the initial magnetization curve. Since $B = \mu_0 \mu_r H$ [5], the relative permeability corresponds to the slope of the curve. Therefore, by taking the derivative of the initial magnetization curve, it can be seen that the relative permeability reaches a maximum value, which occurs in the dynamically changing central region of the curve [6].

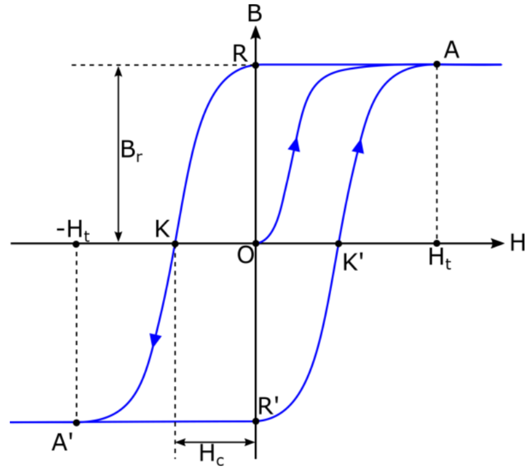

Fig. 2 — Hysteresis curve

As the magnetic field strength depends on the current, the value of the relative permeability is also current-dependent. As the current increases, the relative permeability initially rises rapidly until it reaches its maximum, after which it begins to decrease [7]. The effect of relative permeability on the outer surface impedance results in an inversely proportional relationship between the relative permeability and the screening factor, meaning the screening factor exhibits a minimum value.

C. Measurement of relative permeability

The relative permeability measurement was compiled according to the CCITT Directives Volume IX. [3]. To perform the measurement, a section of the steel tape armour was removed from the cable. The steel tape was then wound around a wooden cylinder, and a conductor was placed at its center. A magnetisation current, denoted as I_m, was passed through the conductor to facilitate the measurement of the steel's relative permeability.

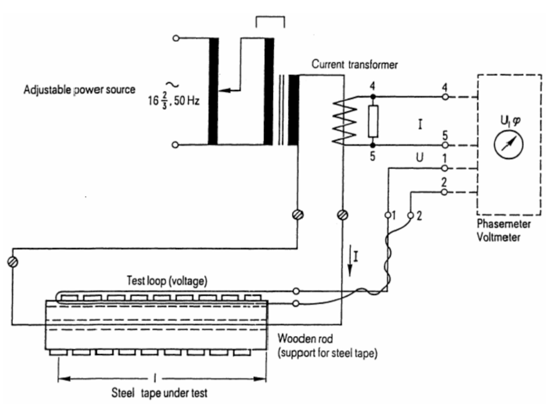

Fig. 3 — Test arrangement [3]

During the measurements the I_m current varied from 0 A to 150 A in steps of 10 A. As it is mentioned above, in order to calculate the value of relative permeability at every point, the U loop voltage between points "1" and "2" in Fig. 3, the value of I_m and the ϕ phase between them should be measured. So as to measure the ϕ phase a digital oscilloscope was used. The current signal was converted into voltage with a current transformer as shown in Fig. 3, then the two voltage signals were connected to the channels of the oscilloscope. In one measurement point, 3 periods of waveform from each channel were captured, and later the ϕ angle was calculated from them.

The geometric parameters for steel tapes from cables "A" and "B" is shown in the table I.

TABLE I: Geometric details of steel tapes

Geometric details		
	A	**B**
D_m [mm]	45	46.65
a [mm]	50	60
b [mm]	1	1.4
n [db]	12.42	6.19

According to (6) the values of relative permeability were calculated and presented on Fig. 4 and 5.

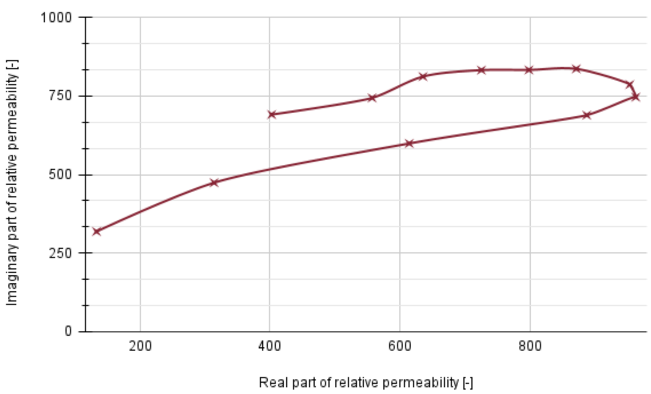

Fig. 4 — Results of relative permeability measurements for cable 'A'

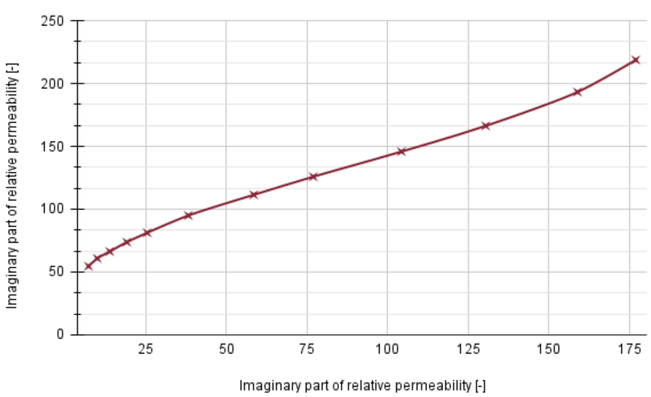

Fig. 5 — Results of relative permeability measurements for cable 'B'

As it is shown on the Fig. 4, the curve reaches its maximum value in the testing range. This is correlated with the fact that the k_i screening factor reaches its minimum value in that range. In contrast, the relative permeability of the steel tape from cable "B" exhibits a different trend, without reaching a maximum value within the testing range. The maximum $Re(\mu_r)$ and the $Im(\mu_r)$ for cable "A" exceed 700, which is a significant difference compared to Fig. 5, where $Im(\mu_r)$ and $Re(\mu_r)$ are just above 175 and 200, respectively.

III. Material Science

In order to determine the reason behind the differing relative permeability values, the steel tapes were thoroughly investigated using tests commonly employed in materials science.

Initially, the investigation focused on the primary alloying elements of the two materials. Energy Dispersive Spectroscopy (EDS) was employed to analyze their composition.

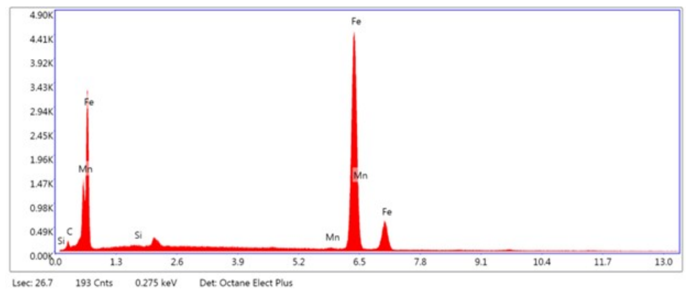

Fig. 6 — Results of EDS for steel from cable 'A'

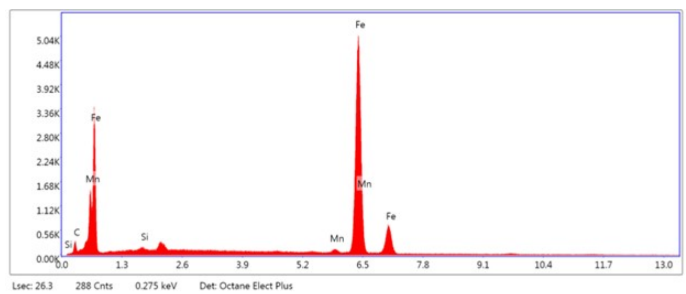

Fig. 7 — Results of EDS for steel from cable 'B'

As shown in Fig. 6 and Fig. 7 the primary alloying elements of the two steel tape armour are identical. Hardness test was also implemented on the two steel test samples. In order to properly measure the hardness of the samples, a *Buehler Indentamet 1100 Vickers* hardness tester was used. The results are shown in Table II.

TABLE II: Hardness testing results

Sample	Result (Vickers)
A	104 HV
B	165 HV

According to scientific papers [8], [9] and [10], the harder the steel, the lower its relative permeability, which correlates with our results.

In order to reveal the causes behind the different hardness values, an optical microscopy test was applied.

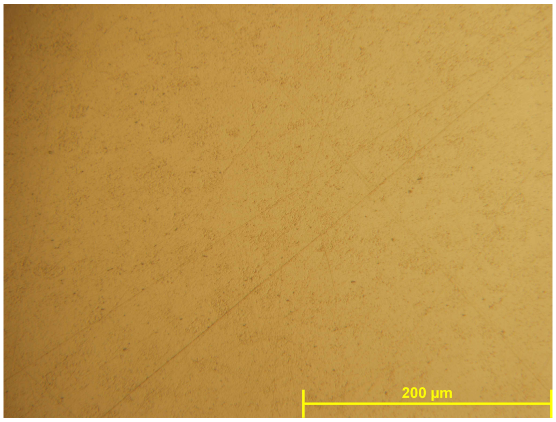

Fig. 8 — Optical microscopy picture for sample ”A”

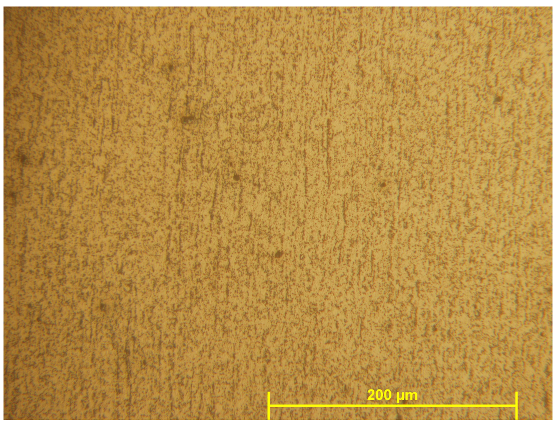

Fig. 9 — Optical microscopy picture for sample ”B”

As shown in Fig. 8, the microstructure of sample ”A” is ferrite-dominated. In contrast, sample ”B” Fig. 9 exhibits a directional, parallel grain structure.

Steel tape armours like these are typically manufactured using cold-rolling techniques [11]. The characteristic microstructure resulting from this process can be observed in Fig. 9. However, the presence of large ferrite grains and the relatively low hardness values in cable ”A” suggest that the material likely underwent heat treatment after the cold-rolling process.

IV. Conclusion

In conclusion, the findings suggest that, in the case of coaxial railway signaling cables, the relative permeability of the steel tape armour plays a key role in determining the inner screening factor. Relative permeability exhibits an inverse proportional relationship with the inner screening factor. Therefore, manufacturing steel tape armours with high relative permeability within the range of typical traction currents is essential.

The second part of this paper clearly indicates that manufacturing techniques have a direct impact on the relative permeability values. To achieve a higher relative permeability, it is advisable to apply heat treatment following the cold-rolling process.

Although this paper provides a clear recommendation regarding the manufacturing process, it is important to note that the conclusions are based on laboratory measurements of two different cables, and the manufacturing techniques were inferred from the observed results. To develop concrete and reliable manufacturing guidelines, the implementation of factory-based testing would be advisable.

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